Forecast daily bike rental demand using time series models

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About Data Analysis Report

This RMarkdown file contains the report of the data analysis done for the project on forecasting daily bike rental demand using time series models in R. It contains analysis such as data exploration, summary statistics and building the time series models.

Data Description: This dataset contains the daily count of rental bike transactions between years 2011 and 2012 in Capital Bikeshare system with the corresponding weather and seasonal information.

Data Source: https://archive.ics.uci.edu/ml/datasets/bike+sharing+dataset

Relevant Paper: Fanaee-T, Hadi, and Gama, Joao, 'Event labeling combining ensemble detectors and background knowledge', Progress in Artificial Intelligence (2013): pp. 1-15, Springer Berlin Heidelberg

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TASK 1: Load and Explore the Data

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```
## Classes 'data.table' and 'data.frame':   731 obs. of  16 variables:
##  $ instant   : num  1 2 3 4 5 6 7 8 9 10 ...
##  $ dteday    : Date, format: "2011-01-01" "2011-01-02" ...
##  $ season    : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ yr        : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ mnth      : num  1 1 1 1 1 1 1 1 1 1 ...
##  $ holiday   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ weekday   : num  6 0 1 2 3 4 5 6 0 1 ...
##  $ workingday: num  0 0 1 1 1 1 1 0 0 1 ...
##  $ weathersit: num  2 2 1 1 1 1 2 2 1 1 ...
##  $ temp      : num  0.344 0.363 0.196 0.2 0.227 ...
##  $ atemp     : num  0.364 0.354 0.189 0.212 0.229 ...
##  $ hum       : num  0.806 0.696 0.437 0.59 0.437 ...
##  $ windspeed : num  0.16 0.249 0.248 0.16 0.187 ...
##  $ casual    : num  331 131 120 108 82 88 148 68 54 41 ...
##  $ registered: num  654 670 1229 1454 1518 ...
##  $ cnt       : num  985 801 1349 1562 1600 ...
##  - attr(*, "spec")=List of 3
##   ..$ cols   :List of 16
##   .. ..$ instant   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ dteday    :List of 1
##   .. .. ..$ format: chr ""
```

```
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_date" "collector"
##   .. ..$ season    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ yr        : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ mnth      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ holiday   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ weekday   : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ workingday: list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ weathersit: list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ temp      : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ atemp     : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ hum       : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ windspeed : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ casual    : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ registered: list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   .. ..$ cnt       : list()
##   .. .. ..- attr(*, "class")= chr [1:2] "collector_double" "collector"
##   ..$ default: list()
##   .. ..- attr(*, "class")= chr [1:2] "collector_guess" "collector"
##   ..$ skip   : num 1
##   ..- attr(*, "class")= chr "col_spec"
##  - attr(*, ".internal.selfref")=<externalptr>
```

```
##     instant          dteday               season            yr        
##  Min.   :  1.0   Min.   :2011-01-01   Min.   :1.000   Min.   :0.0000  
##  1st Qu.:183.5   1st Qu.:2011-07-02   1st Qu.:2.000   1st Qu.:0.0000  
##  Median :366.0   Median :2012-01-01   Median :3.000   Median :1.0000  
##  Mean   :366.0   Mean   :2012-01-01   Mean   :2.497   Mean   :0.5007  
##  3rd Qu.:548.5   3rd Qu.:2012-07-01   3rd Qu.:3.000   3rd Qu.:1.0000  
##  Max.   :731.0   Max.   :2012-12-31   Max.   :4.000   Max.   :1.0000  
##       mnth          holiday            weekday        workingday   
##  Min.   : 1.00   Min.   :0.00000   Min.   :0.000   Min.   :0.000  
##  1st Qu.: 4.00   1st Qu.:0.00000   1st Qu.:1.000   1st Qu.:0.000  
##  Median : 7.00   Median :0.00000   Median :3.000   Median :1.000  
##  Mean   : 6.52   Mean   :0.02873   Mean   :2.997   Mean   :0.684  
##  3rd Qu.:10.00   3rd Qu.:0.00000   3rd Qu.:5.000   3rd Qu.:1.000  
##  Max.   :12.00   Max.   :1.00000   Max.   :6.000   Max.   :1.000  
##    weathersit         temp             atemp              hum        
##  Min.   :1.000   Min.   :0.05913   Min.   :0.07907   Min.   :0.0000  
##  1st Qu.:1.000   1st Qu.:0.33708   1st Qu.:0.33784   1st Qu.:0.5200  
##  Median :1.000   Median :0.49833   Median :0.48673   Median :0.6267  
##  Mean   :1.395   Mean   :0.49538   Mean   :0.47435   Mean   :0.6279  
```

```
##  3rd Qu.:2.000   3rd Qu.:0.65542   3rd Qu.:0.60860   3rd Qu.:0.7302
##  Max.   :3.000   Max.   :0.86167   Max.   :0.84090   Max.   :0.9725
##    windspeed           casual         registered        cnt
##  Min.   :0.02239   Min.   :   2.0   Min.   :  20   Min.   :  22
##  1st Qu.:0.13495   1st Qu.: 315.5   1st Qu.:2497   1st Qu.:3152
##  Median :0.18097   Median : 713.0   Median :3662   Median :4548
##  Mean   :0.19049   Mean   : 848.2   Mean   :3656   Mean   :4504
##  3rd Qu.:0.23321   3rd Qu.:1096.0   3rd Qu.:4776   3rd Qu.:5956
##  Max.   :0.50746   Max.   :3410.0   Max.   :6946   Max.   :8714

##  [1] "instant"     "dteday"      "season"      "yr"          "mnth"
##  [6] "holiday"     "weekday"     "workingday"  "weathersit"  "temperature"
## [11] "feel_temp"   "humidity"    "wind_speed"  "casual"      "registered"
## [16] "total_count"
```

Describe and explore the data

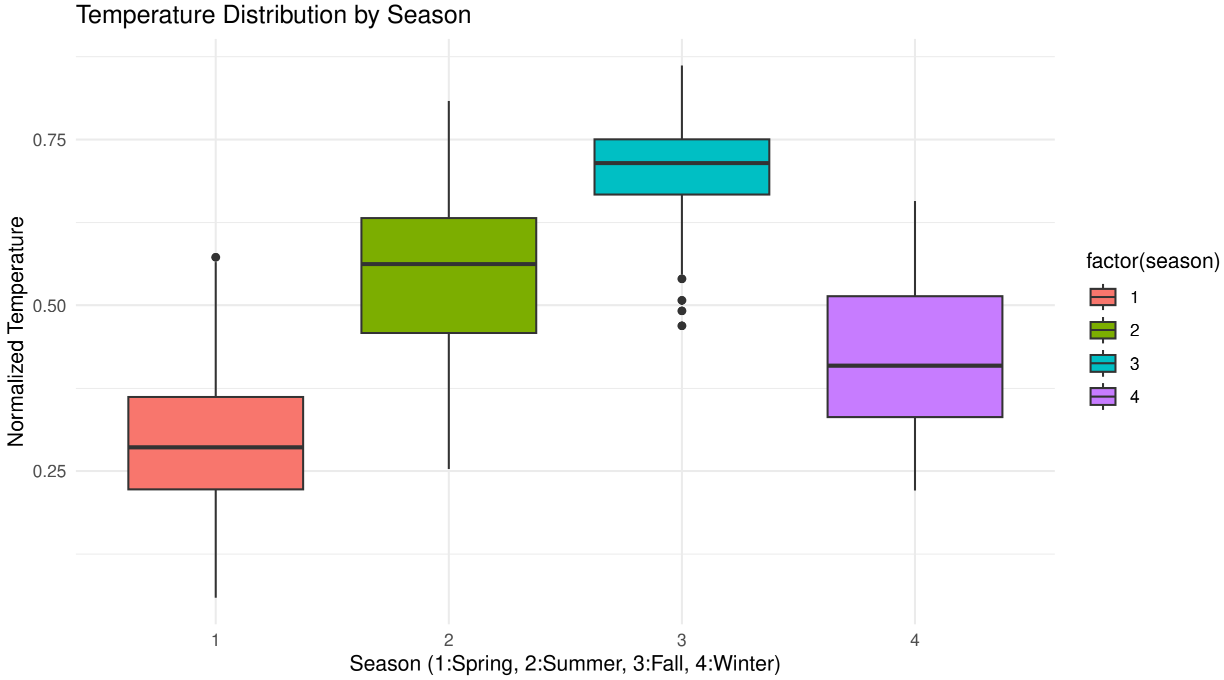
```
##
## === Dataset Overview ===

## Date Range: 2011-01-01 to 2012-12-31

## Total observations: 731

## Missing values:

##     instant      dteday      season          yr        mnth     holiday
##           0           0           0           0           0           0
##     weekday  workingday  weathersit temperature   feel_temp    humidity
##           0           0           0           0           0           0
##  wind_speed      casual  registered total_count
##           0           0           0           0

##
## === Correlation Analysis ===

## Correlation between temperature and total count: 0.627

## Correlation between feeling temperature and total count: 0.631

## Correlation between temperature and feeling temperature: 0.992

##
## === Seasonal Statistics ===

##    season mean_temp median_temp mean_rentals median_rentals
##     <num>     <num>       <num>        <num>          <num>
## 1:      1 0.2977475   0.2858330     2604.133         2209.0
## 2:      2 0.5444052   0.5620835     4992.332         4941.5
## 3:      3 0.7063093   0.7145830     5644.303         5353.5
## 4:      4 0.4229060   0.4091665     4728.163         4634.5

##
## === Monthly Statistics ===

##      mnth mean_temp mean_humidity mean_windspeed mean_rentals
##     <num>     <num>         <num>          <num>        <num>
##  1:     1 0.2364439     0.5858283      0.2063028     2176.339
##  2:     2 0.2992264     0.5674647      0.2156839     2655.298
##  3:     3 0.3905388     0.5884750      0.2226994     3692.258
##  4:     4 0.4699988     0.5880631      0.2344822     4484.900
```

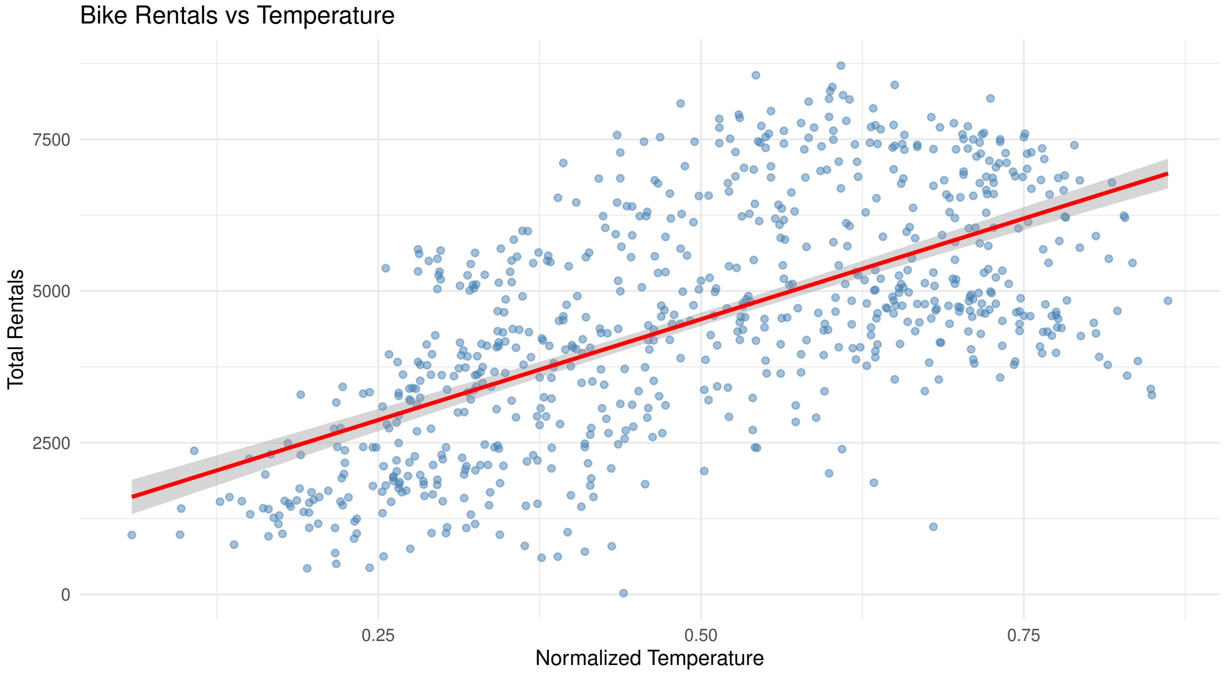
```
##  5:      5 0.5947984     0.6889583      0.1829889     5349.774
##  6:      6 0.6840972     0.5758055      0.1854199     5772.367
##  7:      7 0.7554704     0.5978763      0.1660588     5563.677
##  8:      8 0.7085816     0.6377301      0.1729181     5664.419
##  9:      9 0.6164850     0.7147144      0.1659451     5766.517
## 10:     10 0.4850122     0.6937609      0.1752055     5199.226
## 11:     11 0.3692198     0.6248765      0.1838014     4247.183
## 12:     12 0.3240310     0.6660405      0.1766089     3403.806
```

```
##
## === User Type Analysis ===
```

```
## Correlation between temperature and registered users: 0.54
```

```
## Correlation between temperature and casual users: 0.543
```

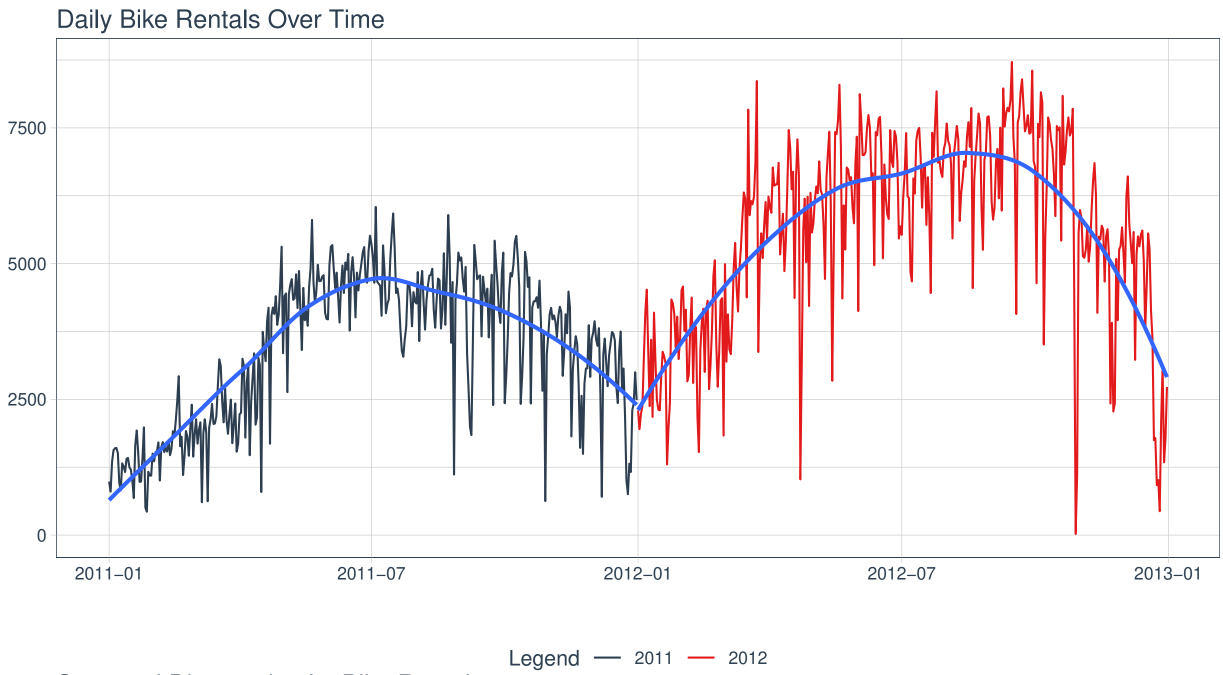


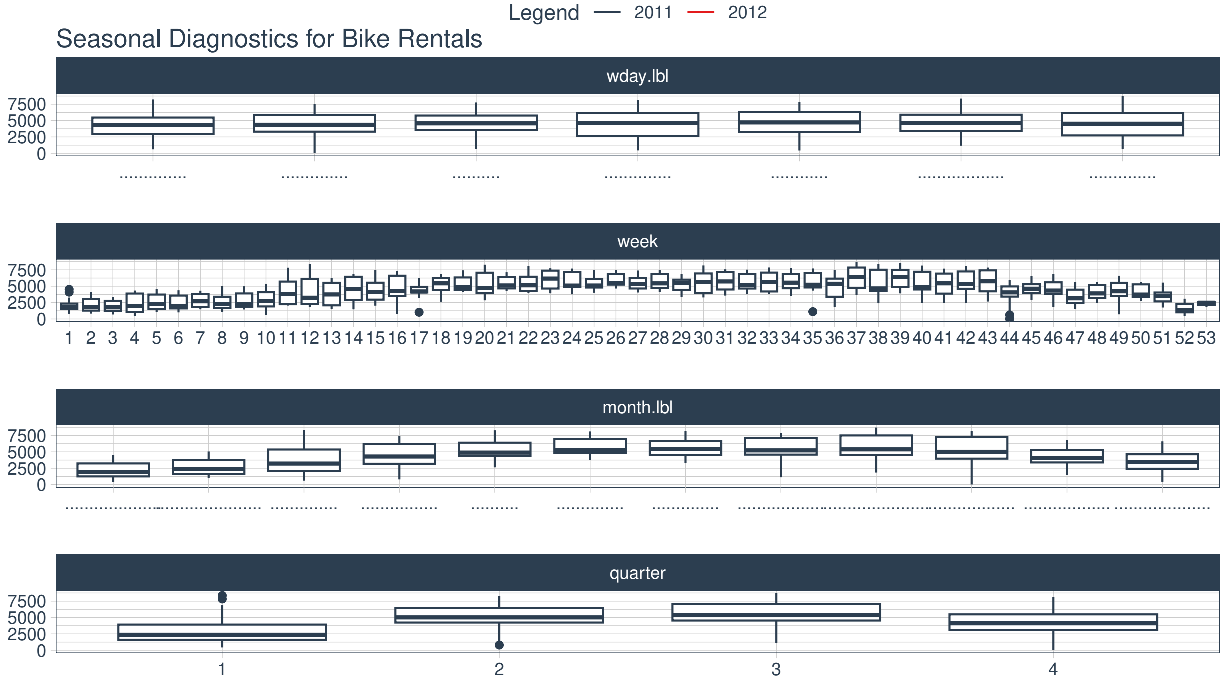

Bike Rentals vs Temperature
Total Rentals
7500
5000
2500
0
0.25
0.50
0.75
Normalized Temperature

==

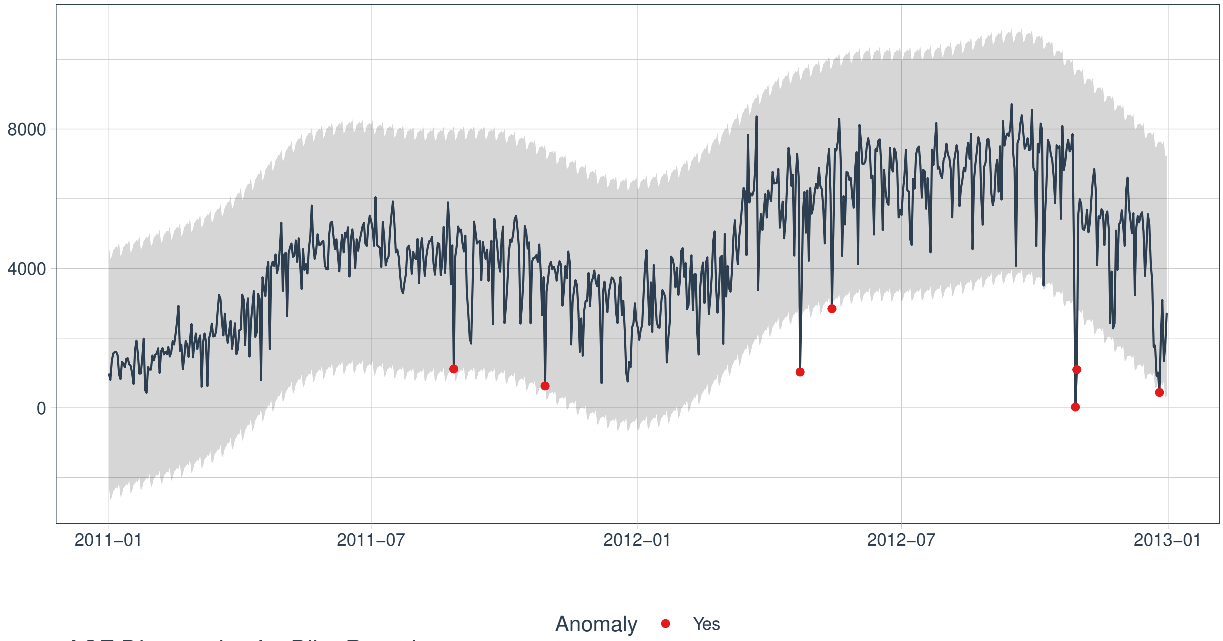
TASK 2: Create Interactive Time Series Plots

==

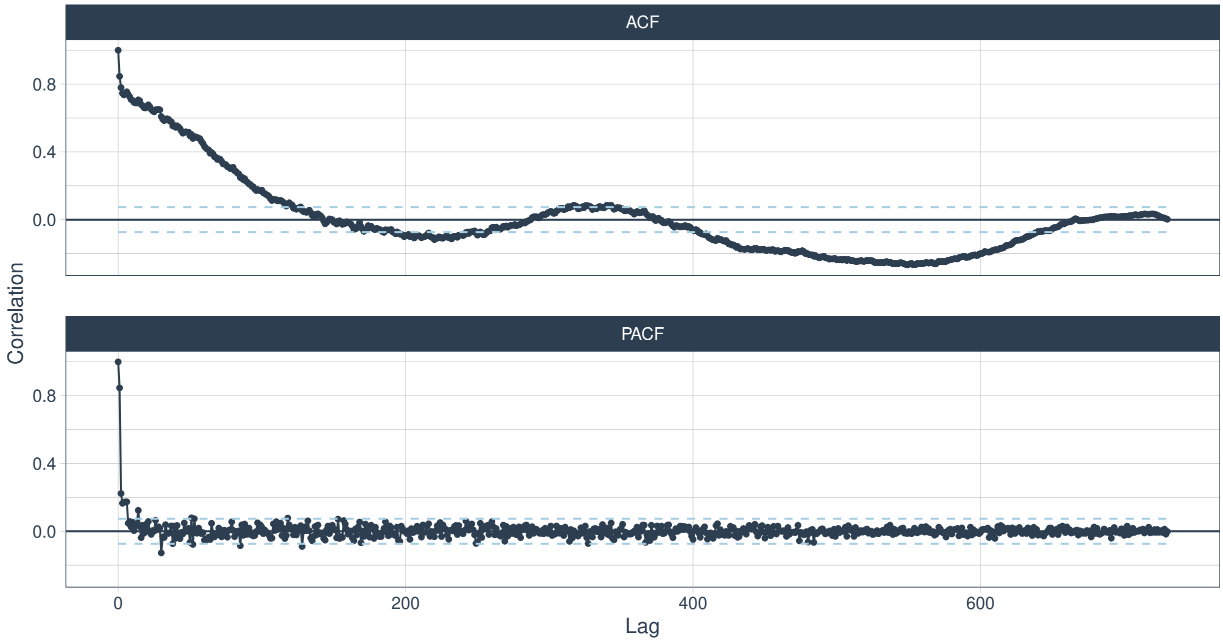




Anomaly Detection in Bike Rentals



ACF Diagnostics for Bike Rentals

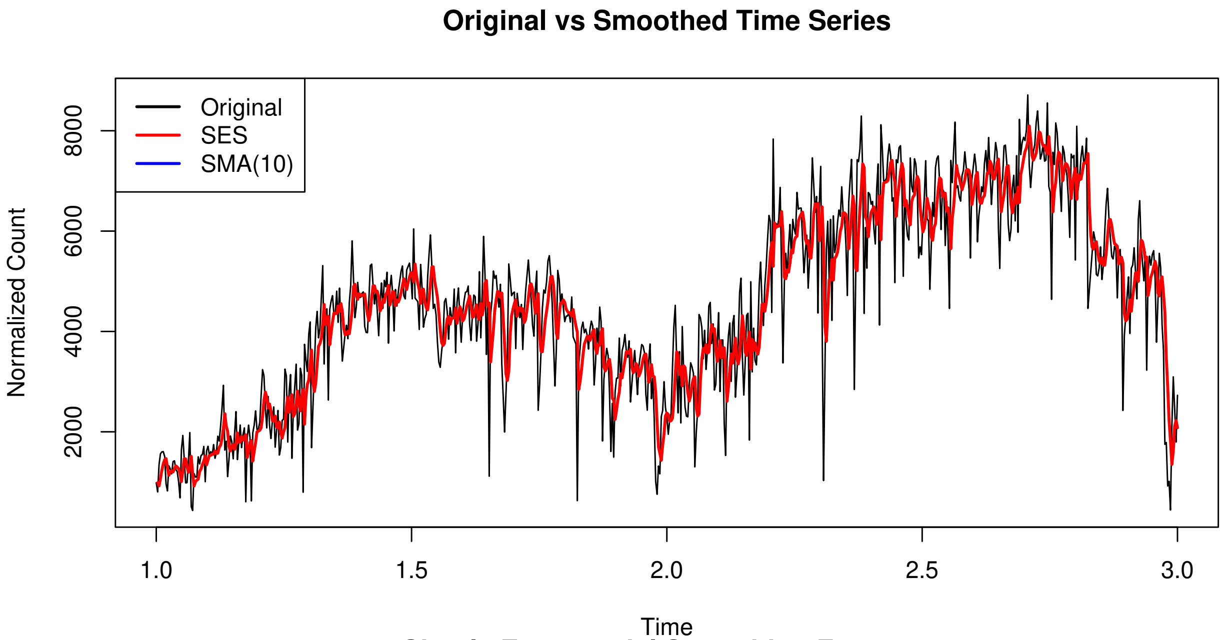


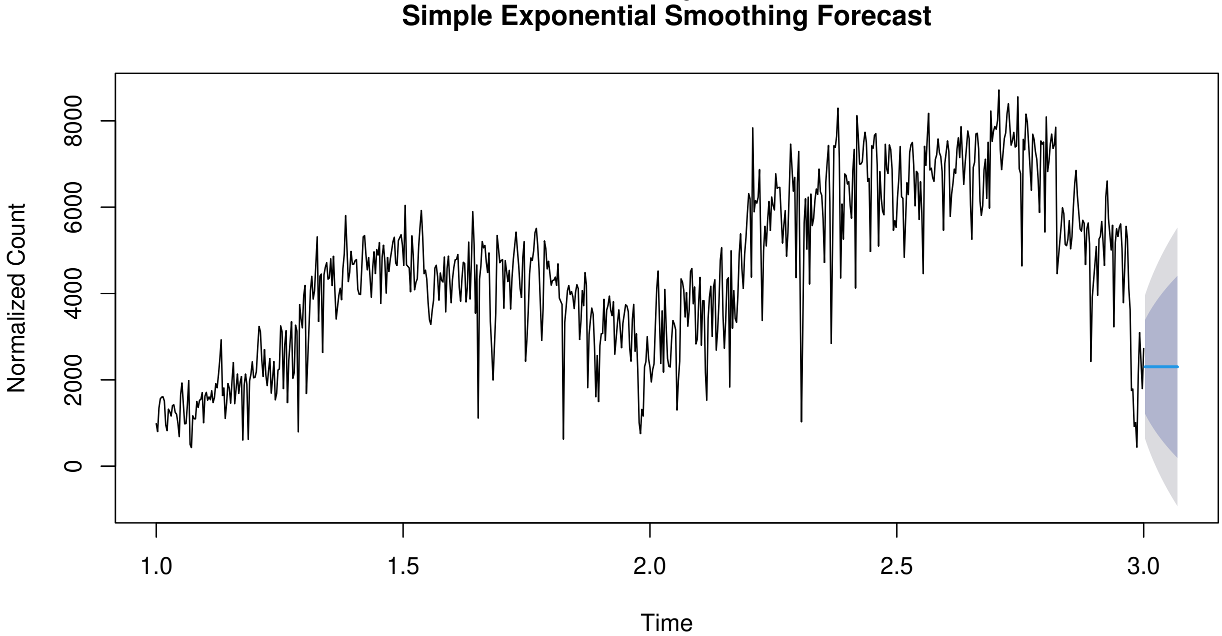
===

TASK 3: Smooth Time Series Data

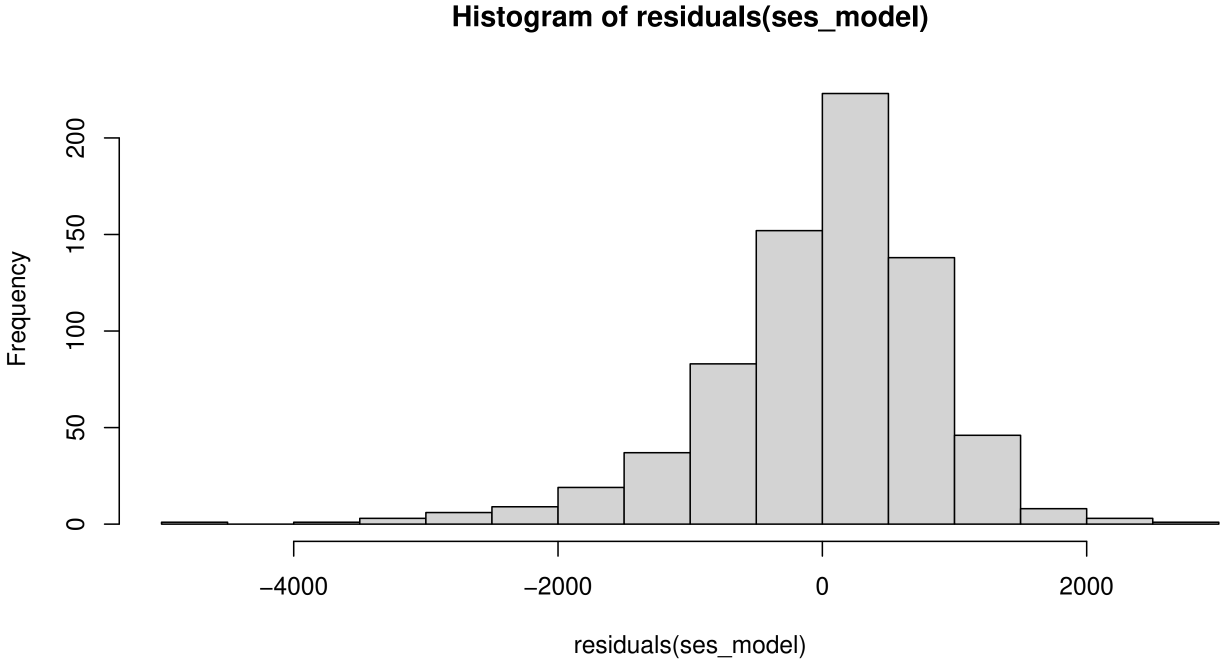
===

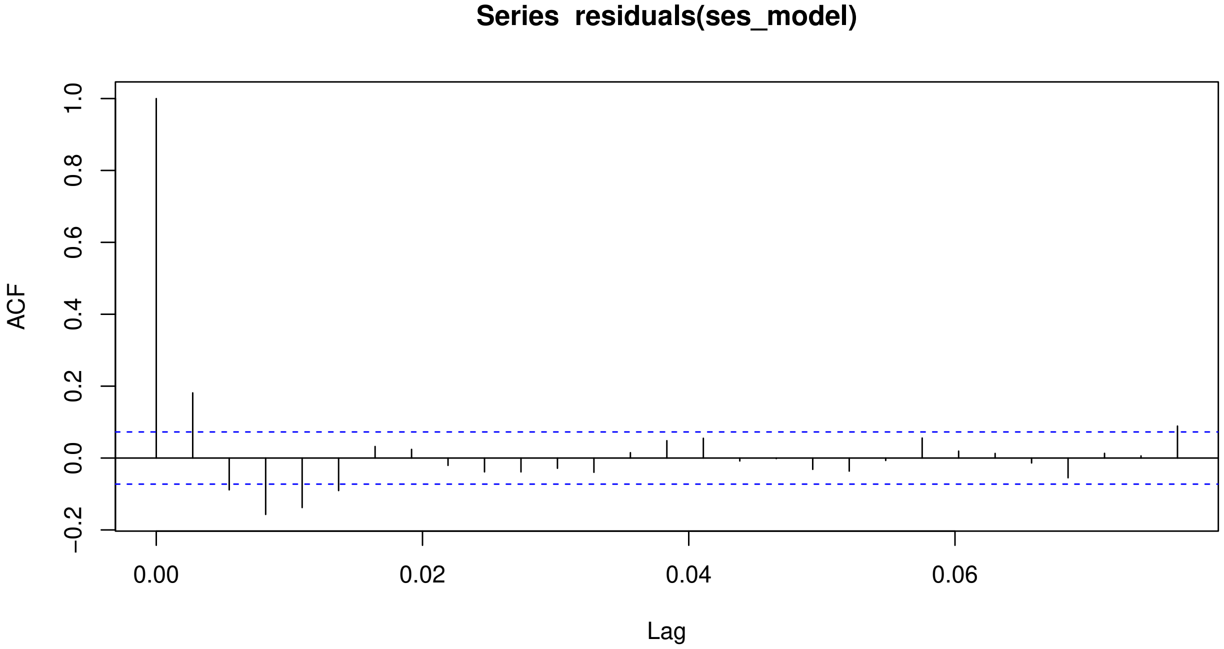
```
## Number of outliers replaced: 30
```






```
##
## === Smoothing Results ===
## Simple Exponential Smoothing alpha: 0.34
```






```
## SES RMSE: 847.1039
```

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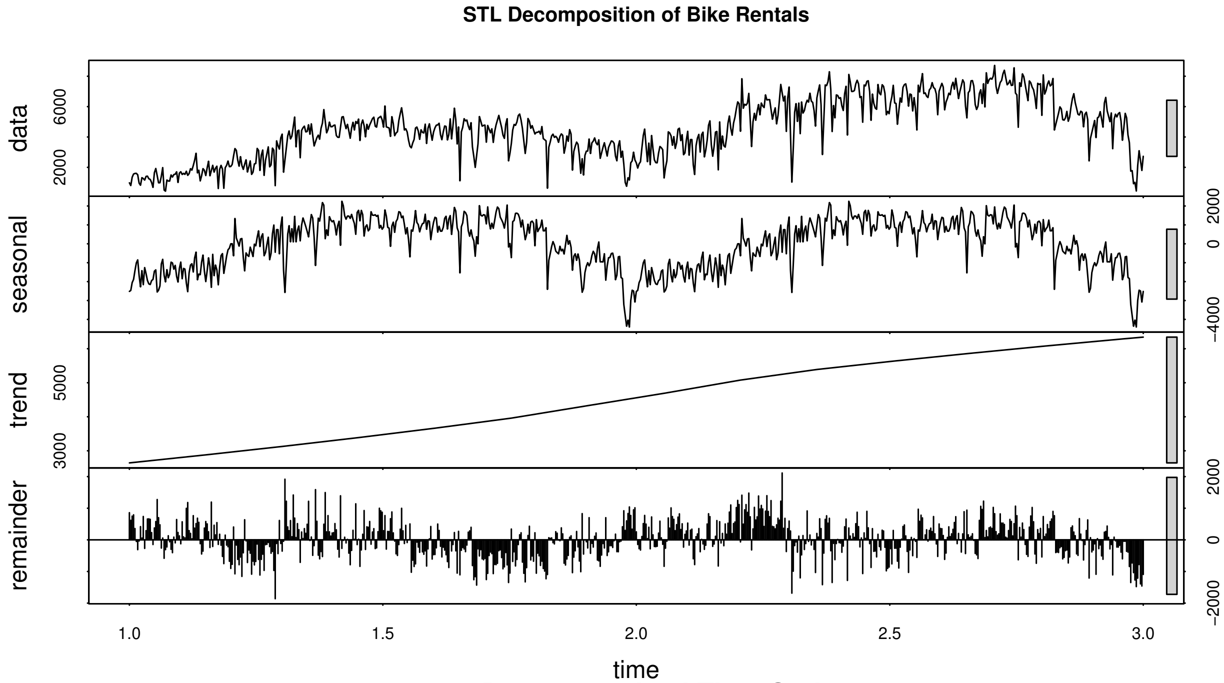
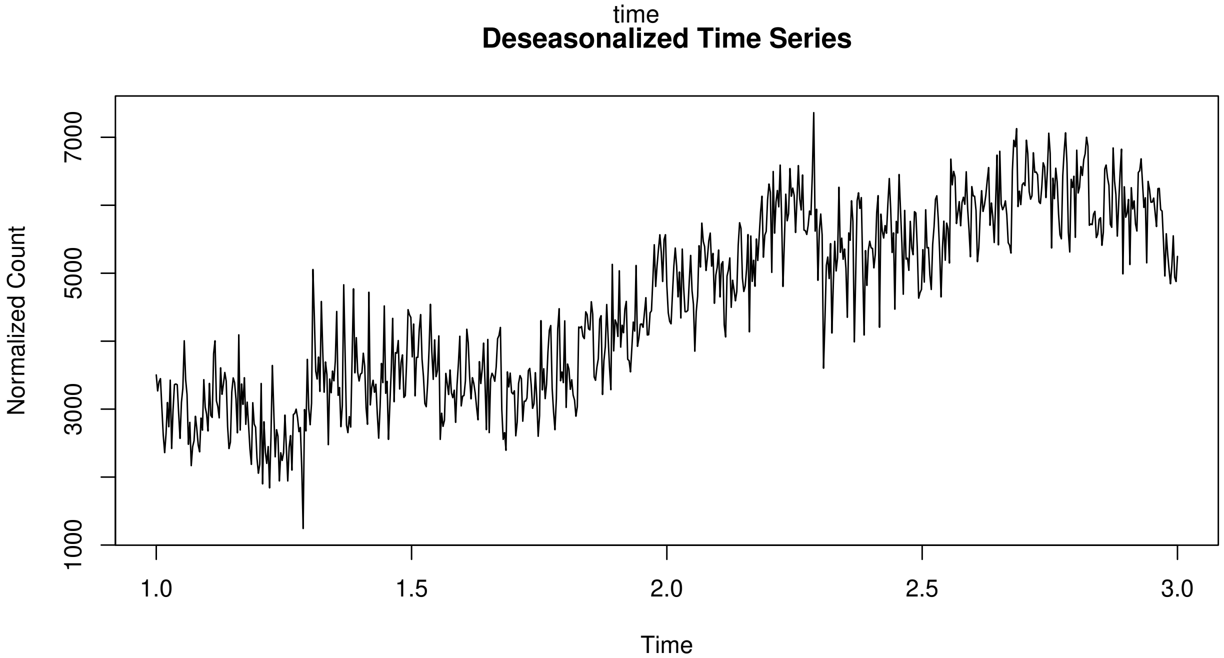
TASK 4: Decompose and Assess Stationarity

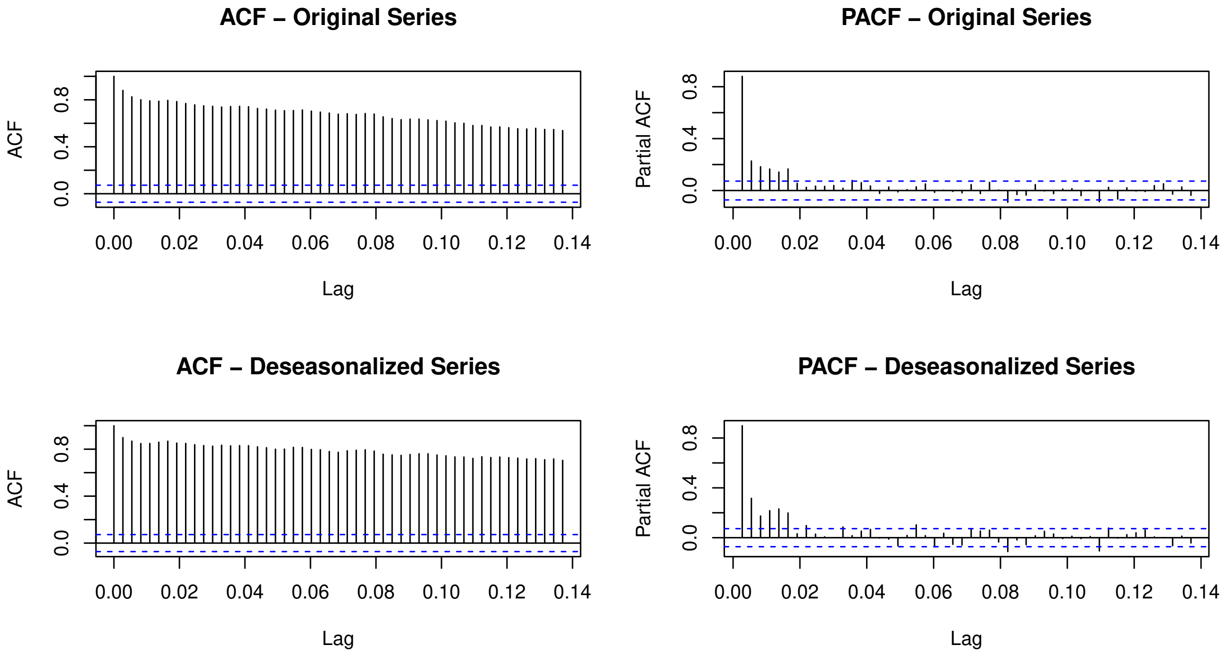
==



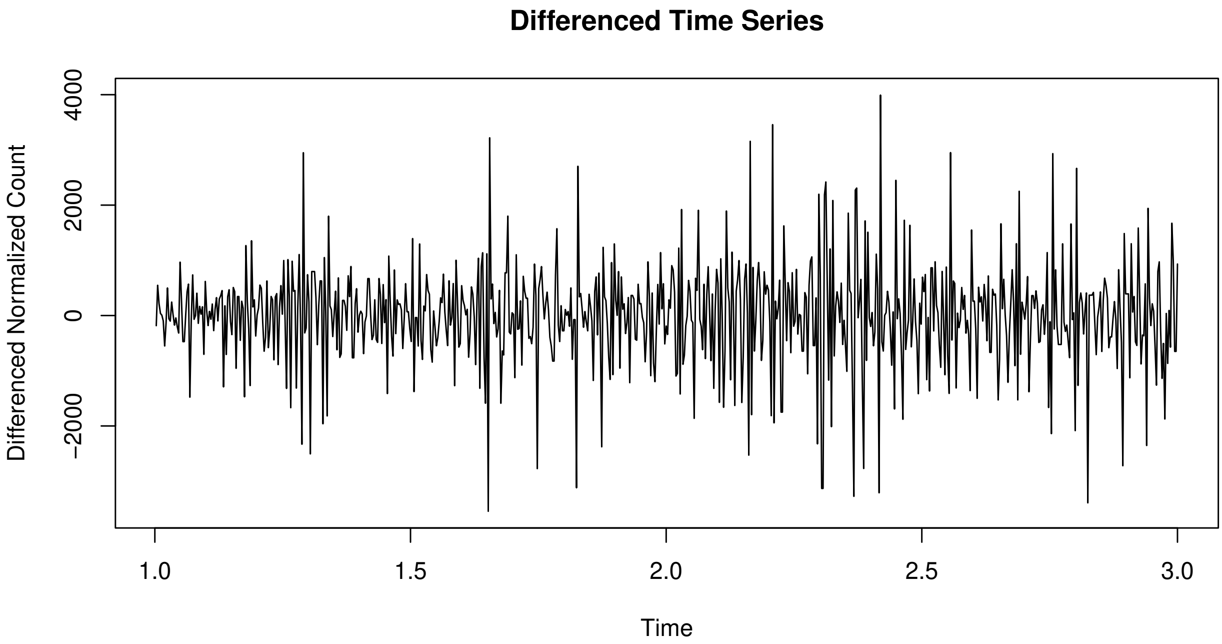



```
##
## === Stationarity Tests ===
##
## ADF Test - Original Series:
```

```
## Test Statistic: -1.4435
## P-value: 0.8138
##
## ADF Test - Deseasonalized Series:
## Test Statistic: -3.4248
## P-value: 0.0495
```




```
##
## === Making Series Stationary with Differencing ===
## Number of differences needed: 1
##
## ADF Test - Differenced Series:
## Test Statistic: -13.2975
## P-value: 0.01
```



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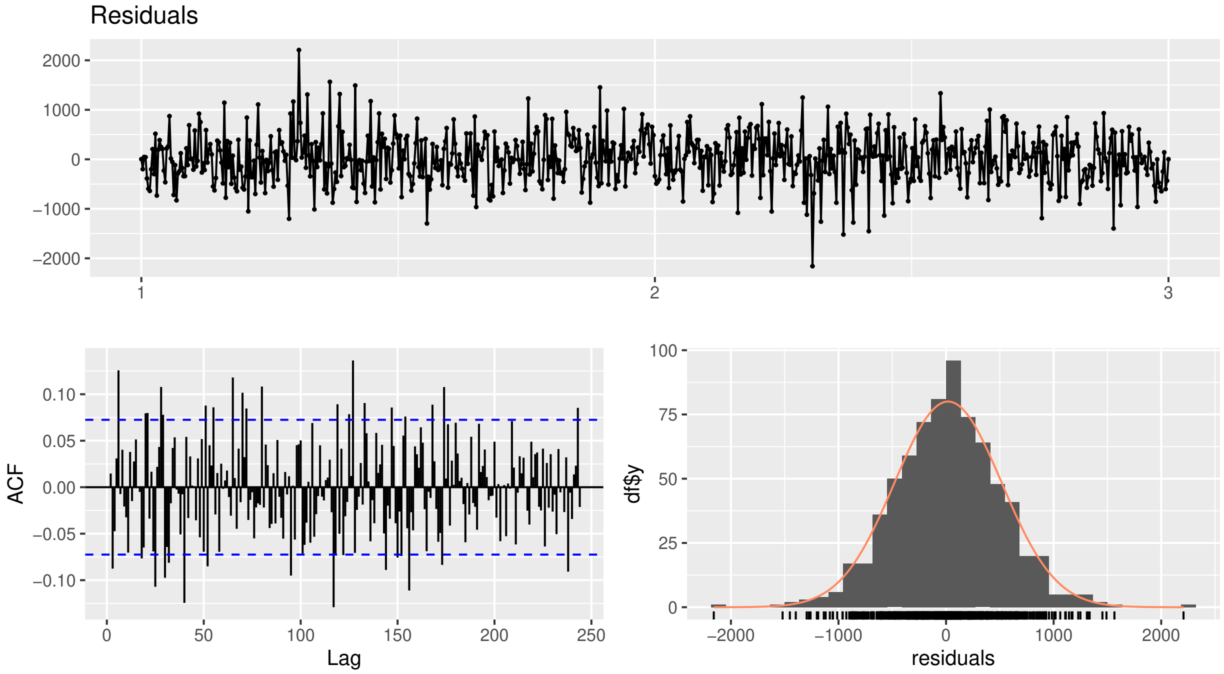
TASK 5: Fit and Forecast using ARIMA Models

==

```
##
## --- Using STLM with auto.arima ---
##
## STLM Model Summary:
##               Length Class          Mode
## stl           2924   mstl           numeric
## model           18   forecast_ARIMA list
## modelfunction    1   -none-         function
## lambda           0   -none-         NULL
## x              731   ts             numeric
## series           1   -none-         character
## m                1   -none-         numeric
## fitted         731   ts             numeric
## residuals      731   ts             numeric
##
## === Residual Diagnostics ===
```

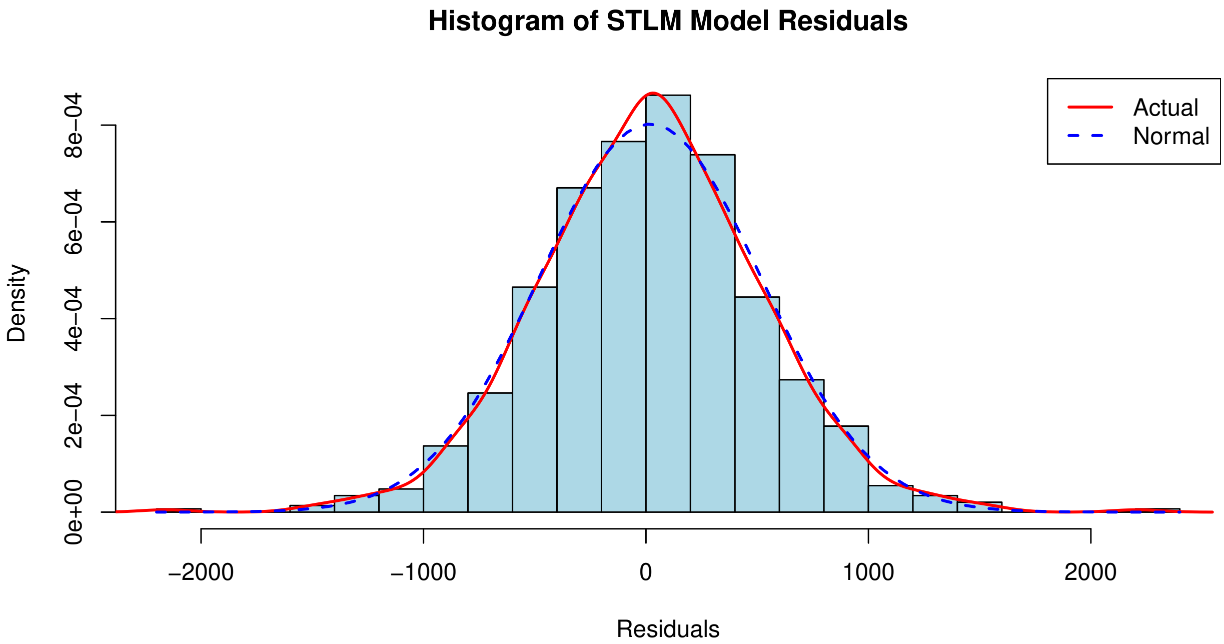



```
##
##  Ljung-Box test
##
## data:  Residuals
## Q* = 339.68, df = 146, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 146
```

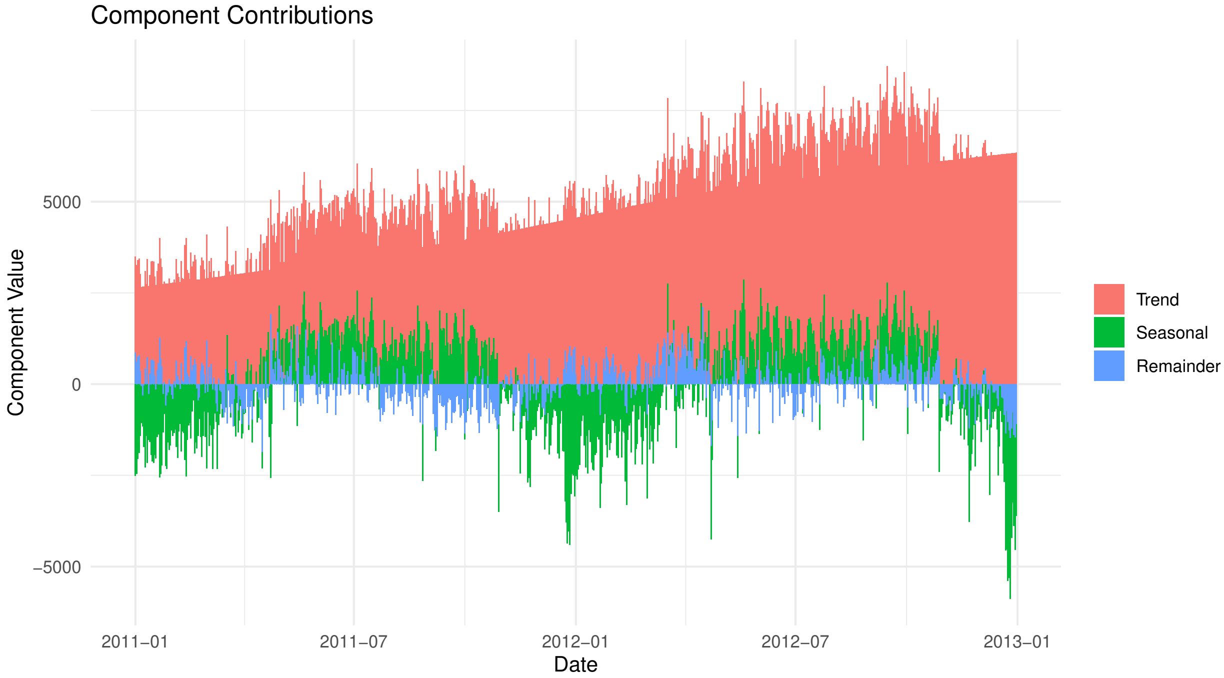
```
##
## Shapiro-Wilk Test for STLM Residuals:
```

```
## Test Statistic: 0.9945
```

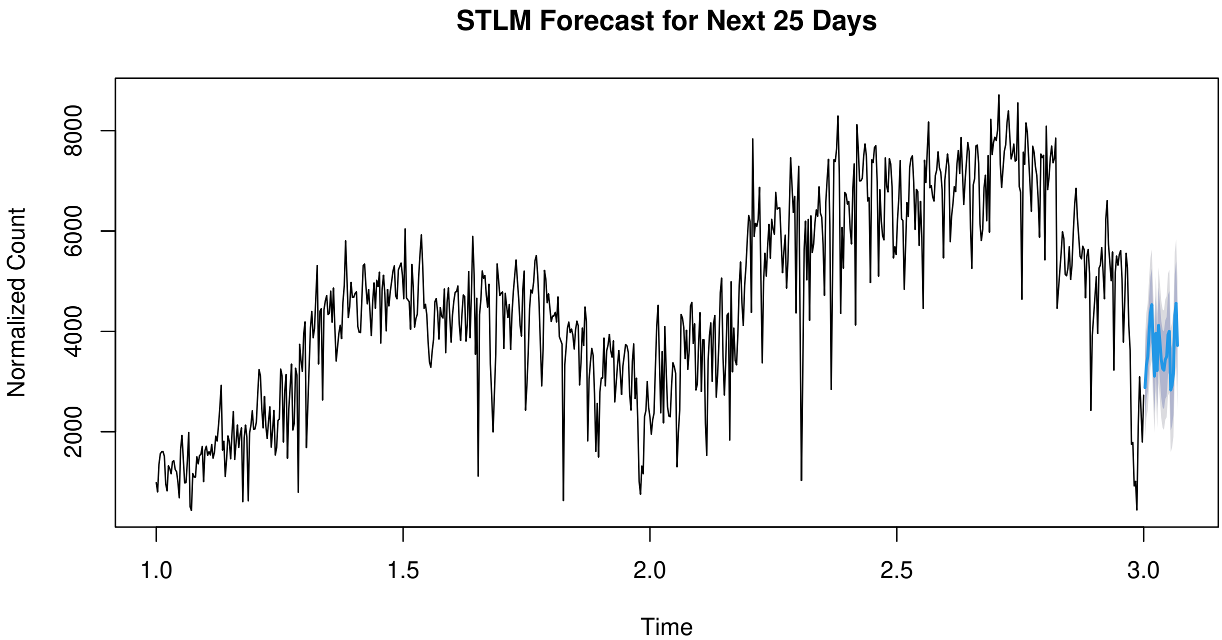
```
## P-value: 0.0098
```




```
##
## Ljung-Box Test:
## Test Statistic: 42.002
## P-value: 0.0028
## Result: Autocorrelation present in residuals
##
## ADF Test - Residuals:
## Test Statistic: -8.0125
## P-value: 0.01
```



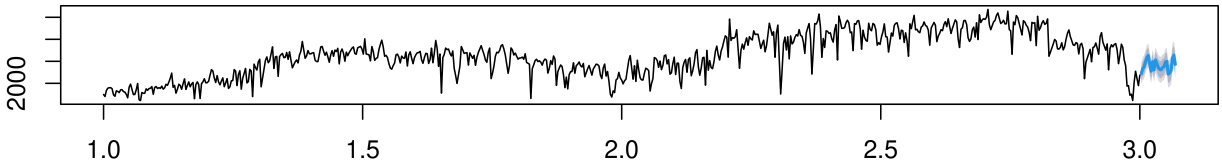

```
##
## === Forecasting ===
```



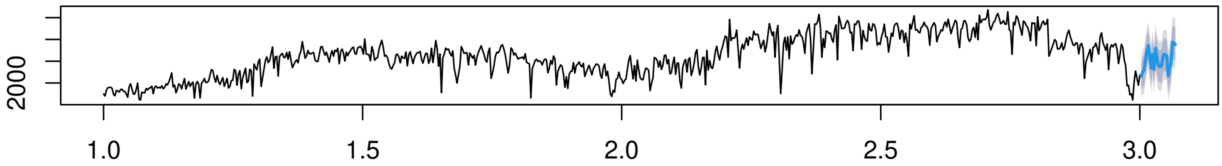

```
##
## Forecast for next 25 days:

##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 3.002740       2879.149 2240.435 3517.863 1902.321 3855.978
## 3.005479       3317.693 2633.851 4001.534 2271.848 4363.538
## 3.008219       3492.850 2795.356 4190.343 2426.126 4559.573
## 3.010959       3961.249 3255.396 4667.101 2881.741 5040.756
## 3.013699       4373.221 3660.259 5086.183 3282.840 5463.603
```

```
## 3.016438       4532.307 3812.600 5252.014 3431.610 5633.004
## 3.019178       3704.288 2977.976 4430.600 2593.490 4815.086
## 3.021918       3106.740 2373.905 3839.576 1985.965 4227.516
## 3.024658       3962.685 3223.389 4701.982 2832.029 5093.341
## 3.027397       3218.628 2472.928 3964.327 2078.179 4359.077
## 3.030137       4123.570 3371.522 4875.618 2973.412 5273.728
## 3.032877       3799.512 3041.169 4557.855 2639.726 4959.297
## 3.035616       3441.954 2677.367 4206.540 2272.620 4611.287
## 3.038356       3259.896 2489.117 4030.674 2081.091 4438.700
## 3.041096       3226.837 2449.915 4003.760 2038.637 4415.037
## 3.043836       3438.779 2655.762 4221.797 2241.258 4636.301
## 3.046575       3496.221 2707.156 4285.286 2289.450 4702.992
## 3.049315       3933.163 3138.096 4728.230 2717.212 5149.113
## 3.052055       4002.605 3201.581 4803.629 2777.544 5227.666
## 3.054795       2830.547 2023.609 3637.484 1596.442 4064.651
## 3.057534       2927.299 2114.492 3740.107 1684.218 4170.381
## 3.060274       3152.552 2333.916 3971.187 1900.557 4404.547
## 3.063014       4316.304 3491.882 5140.726 3055.459 5577.149
## 3.065753       4561.557 3731.388 5391.725 3291.924 5831.190
## 3.068493       3719.809 2883.934 4555.685 2441.448 4998.171

##
## --- Alternative: Direct auto.arima ---

##
## Auto ARIMA Model:

## Series: ts_clean
## ARIMA(1,0,3)(0,1,0)[365] with drift
##
## Coefficients:
##          ar1      ma1      ma2      ma3   drift
##       0.9683  -0.5912  -0.1279  -0.0937  5.7116
## s.e.  0.0224   0.0571   0.0617   0.0576  0.8318
##
## sigma^2 = 986021:  log likelihood = -3042.81
## AIC=6097.63   AICc=6097.86   BIC=6121.05
##
## Training set error measures:
##                    ME     RMSE      MAE       MPE     MAPE      MASE
## Training set 5.853004 697.8113 385.8648 -2.699883 9.189324 0.1694626
##                      ACF1
## Training set -0.003587809

##
## Shapiro-Wilk Test for Auto ARIMA Residuals:

## Test Statistic: 0.838

## P-value: 0
```

STLM Forecast



Auto ARIMA Forecast




```
##
## === Model Comparison ===
##
## STLM Model Accuracy:
##                    ME     RMSE      MAE       MPE     MAPE     MASE
## Training set 18.80439 497.3673 385.0526 -1.938756 12.12249 0.169106
##                      ACF1
## Training set 0.0006157561
##
## Auto ARIMA Model Accuracy:
##                    ME     RMSE      MAE       MPE     MAPE      MASE
## Training set 5.853004 697.8113 385.8648 -2.699883 9.189324 0.1694626
##                      ACF1
## Training set -0.003587809
##
## === Model Selection ===
## STLM AIC: 11145.35
## Auto ARIMA AIC: 6097.63
##
## Best Model: Auto ARIMA (Lower AIC is better)
```